

MANAGEMENT OF HYPERTENSION — CLINICAL-BASED MCQs

1.

A 50-year-old man presents with BP 160/95 mmHg on two occasions. What is the **first-line management**?

- A. Start combination drug therapy immediately
- B. Lifestyle modification
- C. IV antihypertensives
- D. Hospital admission

Answer: B

Rationale: For mild/moderate hypertension without organ damage, begin with **lifestyle modifications** (diet, exercise, weight loss).

2.

A patient with hypertension is advised to reduce salt intake. What is the **maximum recommended daily sodium intake**?

- A. < 2.3 g
- B. < 4 g
- C. < 6 g
- D. < 1 g

Answer: A

Rationale: The **DASH** and WHO guidelines recommend sodium intake < 2.3 g/day to lower blood pressure.

3.

A 45-year-old obese woman with BP 150/90 mmHg asks how exercise helps. Which mechanism best explains this?

- A. Decreases heart rate and vascular resistance
- B. Increases catecholamine activity
- C. Raises cardiac output
- D. Causes water retention

Answer: A

Rationale: **Aerobic exercise** improves cardiac efficiency, reduces heart rate, and decreases systemic vascular resistance.

4.

A patient on hydrochlorothiazide for hypertension reports muscle cramps. What electrolyte imbalance is likely?

- A. Hypernatremia
- B. Hypokalemia
- C. Hyperkalemia
- D. Hypocalcemia

Answer: B

Rationale: Thiazide diuretics increase potassium loss, leading to **hypokalemia**.

5.

A patient develops a persistent dry cough after starting antihypertensive therapy. Which drug is most likely responsible?

- A. Amlodipine
- B. Lisinopril
- C. Losartan
- D. Metoprolol

Answer: B

Rationale: **ACE inhibitors** (e.g., lisinopril) increase bradykinin, causing dry cough.

6.

A patient with hypertension and type 2 diabetes should preferably be started on:

- A. Thiazide diuretic
- B. ACE inhibitor
- C. Beta-blocker
- D. Calcium channel blocker

Answer: B

Rationale: **ACE inhibitors** protect renal function and reduce diabetic nephropathy risk.

7.

A 60-year-old hypertensive man with asthma is prescribed propranolol. What is the nurse's best action?

- A. Administer as ordered
- B. Replace with a cardioselective beta-blocker
- C. Add a diuretic
- D. Stop all antihypertensives

Answer: B

Rationale: **Non-selective beta-blockers** (propranolol) can cause **bronchospasm**; cardioselective options (e.g., metoprolol) are safer.

8.

A hypertensive emergency is defined as BP >180/120 mmHg **with**:

- A. No symptoms
- B. End-organ damage
- C. Dehydration
- D. Chronic cough

Answer: B

Rationale: **Hypertensive emergency** involves **acute target organ damage** (brain, heart, kidney, retina).

9.

A nurse is preparing IV labetalol for a patient with BP 220/130 mmHg and headache. What is the goal of initial BP reduction?

- A. Normalize BP immediately
- B. Reduce mean arterial pressure by $\leq 25\%$ within 1 hour
- C. Reduce diastolic BP to < 60 mmHg
- D. Withhold treatment for 6 hours

Answer: B

Rationale: Rapid BP drop may cause ischemia; gradual $\leq 25\%$ reduction is recommended.

10.

A patient on amlodipine complains of swollen ankles. What causes this side effect?

- A. Sodium retention
- B. Venous dilation and capillary leakage
- C. Reduced cardiac output

D. Hyperkalemia

Answer: B

Rationale: Calcium channel blockers cause **peripheral edema** by arteriolar dilation and fluid shift.

11.

Which antihypertensive is contraindicated in pregnancy?

- A. Methyldopa
- B. Labetalol
- C. ACE inhibitors
- D. Hydralazine

Answer: C

Rationale: ACE inhibitors and ARBs are teratogenic; avoid in pregnancy.

12.

A nurse monitors BP after the first dose of prazosin. Why?

- A. Risk of rebound hypertension
- B. Risk of severe postural hypotension
- C. Risk of hyperkalemia
- D. Risk of bradycardia

Answer: B

Rationale: Alpha-blockers like prazosin can cause **first-dose postural hypotension**.

13.

A 65-year-old with hypertension and benign prostatic hyperplasia (BPH) may benefit from:

- A. Beta-blocker
- B. Alpha-blocker (doxazosin)
- C. ACE inhibitor
- D. Calcium channel blocker

Answer: B

Rationale: Alpha-blockers lower BP and relieve urinary obstruction in **BPH**.

14.

Which dietary measure helps control hypertension?

- A. High saturated fat
- B. Low potassium intake
- C. DASH diet
- D. High sugar intake

Answer: C

Rationale: DASH diet (rich in fruits, vegetables, low-fat dairy) lowers blood pressure.

15.

A patient presents with secondary hypertension. Which cause should the nurse suspect?

- A. Essential hypertension
- B. Renal artery stenosis
- C. Family history
- D. High salt diet

Answer: B

Rationale: Renal artery stenosis is a common cause of **secondary hypertension**.

16.

A patient's BP remains high despite three drugs including a diuretic. This is called:

- A. White coat hypertension
- B. Resistant hypertension
- C. Pseudohypertension
- D. Malignant hypertension

Answer: B

Rationale: Resistant hypertension = uncontrolled BP on ≥ 3 drugs (one must be a diuretic).

17.

Which antihypertensive drug can cause reflex tachycardia and fluid retention?

- A. Hydralazine
- B. Losartan
- C. Metoprolol

D. Amlodipine

Answer: A

Rationale: Direct vasodilators (hydralazine) cause reflex sympathetic activation.

18.

A nurse educates a patient on beta-blockers. Which statement indicates understanding?

- A. "I can stop it once my BP is normal."
- B. "I should not stop it abruptly."
- C. "I can take it only when my BP is high."
- D. "I should double the dose if I miss one."

Answer: B

Rationale: Abrupt withdrawal of beta-blockers causes rebound hypertension and tachycardia.

19.

A patient on losartan develops elevated potassium. What is the nurse's priority?

- A. Stop the drug
- B. Add potassium supplement
- C. Encourage bananas
- D. Increase dosage

Answer: A

Rationale: ARBs can cause hyperkalemia; stop and monitor serum potassium.

20.

Which nonpharmacologic intervention has the greatest impact on reducing BP?

- A. Avoid caffeine
- B. Weight reduction
- C. Increase salt intake
- D. Sleep more

Answer: B

Rationale: Weight loss (even 5–10 kg) significantly lowers BP.

21.

A patient taking thiazides has gout. Why should this drug be changed?

- A. It causes hyperuricemia
- B. It increases urination
- C. It causes hypotension
- D. It raises glucose

Answer: A

Rationale: Thiazides reduce uric acid excretion, leading to gout attacks.

22.

A hypertensive emergency with papilledema is best treated with:

- A. Oral nifedipine
- B. IV labetalol
- C. Oral captopril
- D. IM hydralazine

Answer: B

Rationale: IV labetalol is effective and safe for controlled BP reduction in emergencies.

23.

Which antihypertensive causes gingival hyperplasia?

- A. Amlodipine
- B. Losartan
- C. Methyldopa
- D. Furosemide

Answer: A

Rationale: Calcium channel blockers, especially amlodipine, may cause gum overgrowth.

24.

A patient with chronic kidney disease and hypertension should avoid:

- A. ACE inhibitor
- B. ARB
- C. NSAIDs
- D. Thiazide

Answer: C

Rationale: NSAIDs raise BP and worsen renal function.

25.

A nurse teaches a hypertensive patient to check BP at home. The best time is:

- A. Once per month
- B. Morning before medication
- C. After exercise
- D. Before bedtime

Answer: B

Rationale: Check BP before meds and under resting conditions for accuracy.

26.

A patient on clonidine suddenly stops it and presents with severe hypertension. This is due to:

- A. Drug allergy
- B. Rebound effect
- C. Renal damage
- D. Bradycardia

Answer: B

Rationale: Clonidine withdrawal causes rebound hypertension.

27.

In hypertensive urgency (no organ damage), the best treatment approach is:

- A. IV drugs
- B. Rapid BP reduction
- C. Oral antihypertensives and observation
- D. Hospitalization in ICU

Answer: C

Rationale: Hypertensive urgency can be managed with oral meds and gradual BP control.

28.

A nurse checks BP in both arms; one side shows a large difference. What should be suspected?

- A. Aortic dissection
- B. Stroke
- C. Myocardial infarction
- D. Renal failure

Answer: A

Rationale: Aortic dissection presents with BP discrepancy between arms.

29.

A hypertensive patient with heart failure benefits most from:

- A. ACE inhibitor + diuretic
- B. Alpha-blocker
- C. Calcium channel blocker
- D. NSAIDs

Answer: A

Rationale: ACE inhibitors and diuretics reduce preload and afterload, improving heart failure outcomes.

30.

A patient on spironolactone should be monitored for:

- A. Hypokalemia
- B. Hyperkalemia
- C. Hyponatremia
- D. Hypocalcemia

Answer: B

Rationale: Potassium-sparing diuretics like spironolactone cause **hyperkalemia**.

31.

A patient's BP is well controlled for 6 months. What is the next management step?

- A. Stop all drugs
- B. Gradual dose tapering and continued monitoring
- C. Continue same therapy indefinitely
- D. Change medication

Answer: B

Rationale: Gradual reduction may be attempted under supervision, with close BP monitoring.

32.

A patient on nifedipine reports headache and flushing. What is the cause?

- A. Cerebral edema
- B. Vasodilation
- C. Bradycardia
- D. Hypoglycemia

Answer: B

Rationale: Calcium channel blockers cause **vasodilation**, leading to flushing and headache.

33.

A hypertensive patient is advised to avoid licorice-containing herbal teas. Why?

- A. They increase potassium
- B. They cause sodium and water retention
- C. They cause hypoglycemia
- D. They lower BP excessively

Answer: B

Rationale: Licorice mimics aldosterone, increasing **sodium and water retention** →